

Overview

Lantern Pharma Inc. (NASDAQ: LTRN) is a clinical-stage biopharmaceutical company redefining oncology drug development through artificial intelligence. Its proprietary RADR® platform leverages over 200 billion oncology-specific data points to rapidly identify drug candidates, optimize patient targeting, and de-risk clinical trials—cutting early development time and cost by up to 70% and 80%, respectively. Lantern's pipeline includes three clinical-stage assets: LP-300 (Phase 2) for NSCLC in never-smokers, LP-184 (Phase 1) for DDR-deficient solid tumors, and LP-284 (Phase 1) for aggressive B-cell lymphomas. The company also advances CNS oncology programs via its subsidiary Starlight Therapeutics and is developing novel antibody-drug conjugates (ADCs) in collaboration with global partners. With 11 FDA designations, over 100 issued and pending patents, and partnerships spanning Johns Hopkins, MD Anderson, and Bielefeld University, Lantern is positioned to unlock value through precision-driven clinical execution and strategic collaborations—all backed by \$19.7 million in cash and a lean operating structure.

Analyst Coverage

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Investment Highlights

AI-Driven Drug Development Platform with Transformational Efficiency

- Proprietary AI engine – RADR® – integrates 200B+ oncology-focused datapoints to streamline and de-risk oncology drug development.
- Launched 12 programs in 24 months, reducing time to Phase 1 by ~70% and early-stage development costs by ~80%.
- Patient response predictions validated at >88% accuracy; AI powers biomarker-guided patient selection and combination strategies.

Diverse Pipeline Anchored by Three Clinical-Stage Oncology Programs

- **LP-300 (Phase 2):** Targets relapsed NSCLC in never-smokers—a \$4B+ global market—with demonstrated >2x improvement in median survival for subgroups in prior trials. Currently enrolling patients in the multinational Harmonic™ trial.
- **LP-184 (Phase 1):** Designed for DDR-deficient solid tumors including pancreatic, bladder, TNBC, and glioblastoma. Fast Track and Orphan Drug designations; monotherapy and combination trials underway.
- **LP-284 (Phase 1):** Novel acylfulvene analog for aggressive B-cell lymphomas (MCL, HGBL). Orphan Drug designation in place; potent single-agent and combination activity in preclinical models, including Ibrutinib-resistant tumors.

Multiple Platform Expansion Opportunities

- Wholly owned Starlight Therapeutics subsidiary advancing STAR-001 for CNS and pediatric brain cancers; holds Rare Pediatric Disease and Orphan Drug designations.
- Cryptophycin-loaded Antibody-Drug Conjugates (ADCs) showing sub-nanomolar potency; highly stable constructs with superior drug-to-antibody ratios vs. MMAE benchmarks.
- Early-stage pipeline includes ER+ breast cancer program & multiple RADR®-identified candidates with combination & licensing potential.

Strategic Positioning with Strong IP and Regulatory Tailwinds

- 11 designations across the pipeline, including multiple Fast Track, Orphan Drug, and Rare Pediatric Disease designations.
- 100+ issued and pending patents across RADR®, LP-300, LP-184, and LP-284—extending exclusivity through 2044.
- Modular, reusable AI infrastructure allows for rapid asset generation without redundant infrastructure or fixed costs.

Capital-Efficient Model with a Clear Path to Value Creation

- \$19.7 million in cash (as of March 31, 2025); fully diluted share count of ~12.1 million.
- Active collaborations with Johns Hopkins, Bielefeld University, and others provide validation and external development pathways.
- Multiple upcoming catalysts with Phase 1/2 trial readouts for LP-300 and LP-284, Phase 1a completion for LP-184, and ADC IND-enabling studies in progress.

Recent / Upcoming Milestones

- ✓ Granted FDA Fast Track Designation for LP-184 in GBM and TNBC
- ✓ Multiple clinical readouts for: Phase 2, LP-300 Trial, Phase 1a LP-184 & LP-284 Trial
- ✓ Secured FDA IND clearance for Phase 1b/2 trial of LP-184 in NSCLC and TNBC
- ✓ Potential spinout of Starlight Therapeutics
- ✓ Expand & commercialize RADR® and launch initial modules as open-source AI agents

Collaborations

World-Class Academic and Research Institutions



Biopharma Collaborations



Diverse & Unique AI Driven Pipeline of Drug Programs

Lantern Pharma (NASDAQ: LTRN)								
Lead Program	Indication	Discovery	Preclinical	Phase I	Phase II	Orphan Designation	Rare Pediatric Disease	Fast Track
LP-300	Non-Small Cell Lung Cancer for Never Smokers							
LP-184	Recurrent Advanced Solid Tumors (Pancreatic, TNBC, Bladder, Lung Cancers)							
LP-284	Recurrent Non-Hodgkin's Lymphomas (Mantle cell, Double-hit lymphomas)							
ADC	Select Solid Tumors							
Starlight Therapeutics (Wholly Owned Subsidiary)								
STAR-001 (LP-184 for CNS and Brain Cancers Only)	First Recurrent Glioblastoma in adults							
	Newly Diagnosed MGMT Unmethylated Glioblastoma (investigator led trial)							
	Phase 1a monotherapy including ATRT, DIPG and Medulloblastoma							
	Phase 1b combination select pediatric CNS cancers							
RADR® Collaborations								
Elraglusib owned by - Actuate Therapeutics	Multiple Solid Tumors					Collaboration partner		
TTC-352 owned by - TTC Oncology	ER+ Breast Cancers					Collaboration partner		
XCE853 owned by - Oregon Therapeutics	Protein Disulfide Isomerase (PDI) Inhibitor					Collaboration partner		
ADC	Cryptophycin Conjugate for Solid Tumors					Collaboration partner		

Value Proposition

Lantern Pharma represents a compelling investment opportunity at the intersection of artificial intelligence and precision oncology, offering an asymmetric upside profile underpinned by clinical-stage assets, a capital-efficient development model, and a validated, proprietary AI engine in RADR®. By leveraging over 200 billion oncology-specific data points, RADR® enables rapid and predictive drug development—cutting both cost and time to Phase 1 by approximately 80% and 70%, respectively. This data-centric platform has already generated three differentiated clinical programs with blockbuster market potential and multiple FDA designations, including Fast Track and Orphan Drug status.

At the core of the value proposition are Lantern's three lead assets: LP-300 (Phase 2), a disulfide modulator targeting relapsed NSCLC in never-smokers with an \$4B+ addressable market; LP-184 (Phase 1), an acylfulvene analog targeting DDR-deficient solid tumors including pancreatic, TNBC, and glioblastoma with over \$10B in combined opportunity; and LP-284 (Phase 1), a synthetic lethality-driven candidate for aggressive non-Hodgkin's lymphomas. Each candidate is guided by RADR® insights and backed by translational data demonstrating superiority over standard-of-care and resistance-breaking potential. In parallel, Lantern's Starlight Therapeutics subsidiary and ADC initiatives offer additional near- and long-term catalysts, with first-in-human trials and platform monetization pathways on the horizon.

A key differentiator for Lantern is its ability to generate and validate high-potential drug programs at a fraction of traditional biotech costs, with 12 programs launched in just two years and an average Phase 1 readiness cost of ~\$2 million per asset. The company's IP moat, with 100+ issued and pending patents, strengthens its commercial defensibility, while collaborations with top-tier academic and biopharma partners enhance optionality for out-licensing and joint development. With a lean operating footprint, \$19.7 million in cash, and a fully diluted share count of just ~12 million, Lantern is well-positioned to deliver shareholder value through data-driven execution, targeted partnerships, and strategic advancement of high-impact oncology therapeutics.

For investors seeking early exposure to a platform-based oncology innovator capable of repeatedly generating de-risked, AI-guided assets with high commercial potential, Lantern Pharma offers a differentiated and scalable path to value creation in one of the highest-need therapeutic areas.

Market Data

Lantern Pharma Inc. NASDAQ: LTRN

Price ¹	\$3.16
52 Week Range ¹	\$2.55-\$6.12
Mkt Cap ¹	\$34.1M
Shares Out ¹	10.8M
Float	9.3M
Cash (mrq)	\$19.7M
Insider Ownership	15%
Inst. Ownership	22%

¹ as of June 30, 2025

lanternpharma.com

Legal Counsel:
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Auditor:
EISNERAMPER LLP